Retail Sales Performance, Customer Behaviour & Profitability Analysis Using Python and Power BI

1. Project Overview and Objective

This project focuses on analysing retail sales data using Python and Power BI to understand sales performance, customer behaviour, and business profitability.

- Collected and explored retail sales data to identify trends, patterns, and business performance.
- Performed data cleaning, preprocessing, and transformation using Python (Pandas and NumPy).
- Conducted Exploratory Data Analysis (EDA) to identify sales trends, customer purchasing patterns, and product performance.
- Used visualizations to analyse revenue, profit, discounts, product categories, regional sales, and customer segments.
- Developed interactive Power BI dashboards to present key performance indicators (KPIs) and actionable business insights.

The main objectives of this project are:

- To demonstrate data cleaning, preprocessing, and transformation techniques using Python.
- To analyse retail sales performance and identify trends across different products, categories, regions, and time periods.
- To understand customer purchasing behaviour and identify valuable customer segments.
- To evaluate business profitability by analysing sales, profit, discounts, and other key performance metrics.
- To develop interactive Power BI dashboards that help businesses make data-driven decisions and improve overall performance.

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2. Data Sources

Source Description and Timeline:

- Source: https://www.kaggle.com/datasets/vigneshpgd/indian-retail-sales-analytics-dataset
- Dataset Size: 15,000 records and 22 columns in the original dataset. A 15,000-records and 10 columns sample is used for the project.

- Data Format: Structured data containing customer details, product information, sales transactions, discounts, and profitability metrics.
- Timeline: Based on the Order_Date column in the dataset. Specify the actual date range after checking the minimum and maximum dates.
- Domain: Retail Analytics

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3. Problem Statement

- To analyse sales performance across different regions, product categories, and time periods to identify top-performing areas and sales trends.
- To evaluate overall business profitability by examining revenue, profit margins, discounts, and product performance.
- To understand customer purchasing behaviour and identify customer segments based on their buying patterns.
- To analyse the impact of discounts on sales and profitability to identify opportunities for better pricing strategies.
- To identify high-performing and low-performing products, categories, and regions to support business growth.
- To provide meaningful business insights through Python-based analysis and interactive Power BI dashboards to support effective, data-driven decision-making.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Order_ID	Object (String)	Unique identifier for each sales order
Order_Date	Object (String)	Date on which the customer placed the order
Customer_ID	Object (String)	Unique identifier assigned to each customer
Region	Object (Categorical)	Geographical region where the sales transaction occurred
Category	Object (Categorical)	Main category of the product sold
Subcategory	Object (Categorical)	Specific subcategory of the product sold
Quantity	Float64 (Numeric)	Number of product units purchased in each order

Unit_Price_INR	Float64 (Numeric)	Selling price of one product unit in INR
Sales_INR	Float64 (Numeric)	Total sales amount before deducting discounts
Discount_Amount_INR	Float64 (Numeric)	Discount amount applied to the sales transaction
Net_Sales_INR	Float64 (Numeric)	Final sales amount after deducting discounts
Cost_INR	Float64 (Numeric)	Total cost associated with the products sold
Profit_INR	Float64 (Numeric)	Profit earned from each sales transaction
Returned	Object (Categorical)	Indicates whether a product has been returned
Customer_Satisfaction	Float64 (Numeric)	Customer satisfaction rating for the shopping experience

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5. Tools & Technologies

Python: Data cleaning, data preprocessing, data transformation, Exploratory Data Analysis (EDA), and statistical analysis using Pandas, NumPy, Matplotlib, and Seaborn.

Power BI: Data modelling, data transformation, DAX calculations, KPI development, data visualization, and interactive dashboard creation.

Google Colab: Used for executing Python code and performing data analysis.

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6. Data Pre-Processing (python/ google colab)

Tasks Performed:

Data Cleaning & Transformation:

- Checked dataset structure, column names, data types, and dimensions using Pandas.
- Identified and handled missing and null values in columns such as Region, Category, Quantity, and Customer_Satisfaction.
- Checked for duplicate records and removed duplicates wherever necessary.
- Converted Order_Date from object format to datetime format.
- Corrected inappropriate data types and standardized categorical values.
- Checked numerical columns for inconsistent values and potential outliers.
- Performed data transformation to ensure consistency and accuracy.
- Created calculated fields to support sales, discount, and profitability analysis.
- Verified the cleaned dataset before proceeding with Exploratory Data Analysis (EDA).

Tools & Libraries Used:

- Python
- Google Colab
- Pandas
- NumPy

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7. Data Modelling and DAX (Power BI)

Data Model:

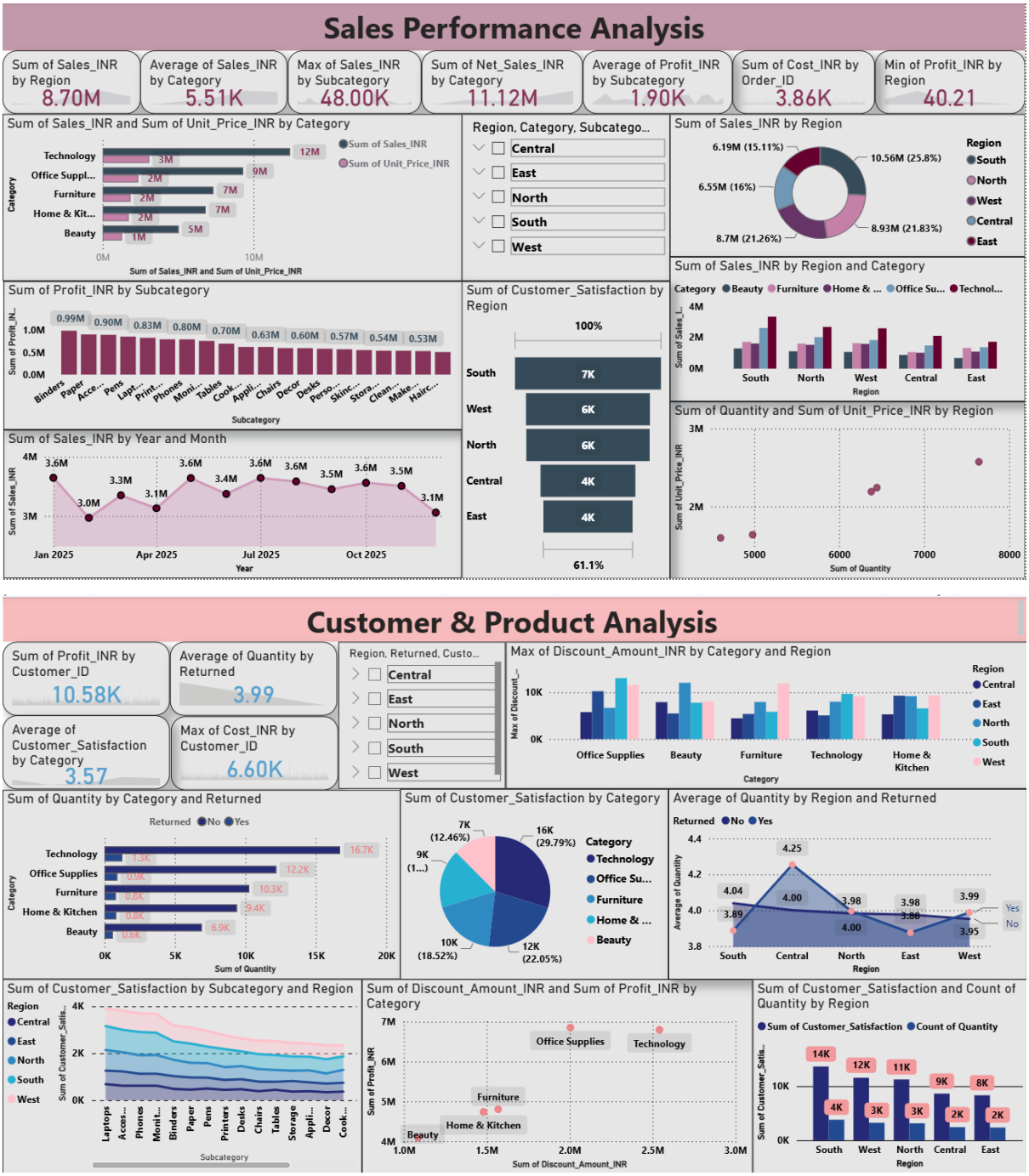
- Imported the cleaned retail sales dataset into Power BI for data modelling and visualization.
- Established relationships between tables wherever required.
- Defined appropriate relationship cardinality (One-to-Many / Many-to-One).
- Created lookup tables where necessary to organize customer, product, and sales information.
- Ensured proper relationships between tables to maintain data consistency and accuracy.

DAX Calculations:

- Created DAX measures to calculate Total Sales, Total Profit, Total Orders, and Total Quantity Sold.
- Calculated Average Order Value (AOV) to understand customer spending patterns.
- Developed profit margin and discount percentage measures to evaluate business profitability.
- Created KPI measures to track sales performance and customer behaviour.
- Used calculated columns and measures to support interactive dashboard analysis.

Tools Used: Power BI

SCREENSHOT OF DASHBOARD /REPORT



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9. Insights & Conclusions

- **Key Findings:**

1. Sales Performance Findings

- The South region recorded the highest sales, contributing approximately ₹10.56 million (25.8%) of total sales.
- The East region recorded the lowest regional sales, contributing approximately ₹6.19 million (15.11%).

- Technology emerged as the highest-performing product category, generating approximately ₹12 million in sales.
- Beauty recorded comparatively lower sales than the other product categories.
- Monthly sales remained relatively stable throughout 2025, with February and December showing comparatively lower sales.

2. Profitability Findings

- Binders generated the highest profit among the displayed subcategories, contributing approximately ₹0.99 million.
- Paper and Accessories were also among the higher profit-generating subcategories.
- Profitability varied across product subcategories, indicating differences in their contribution to overall business performance.
- Discount distribution varied across regions and product categories, highlighting the importance of monitoring promotional strategies.

3. Customer Behaviour Findings

- The overall average customer satisfaction score displayed in the dashboard is approximately 3.57.
- Customer purchasing quantities varied across different regions.
- Technology recorded the highest quantity of non-returned products among the displayed categories.
- Returned product quantities were comparatively lower than non-returned quantities across the displayed categories.
- Customer profitability varied, indicating differences in the profit contribution of individual customers.

4. Regional and Product Performance Findings

- South was the leading region in overall sales performance.
- Technology contributed significantly to retail sales compared with other product categories.
- Regional sales and product category performance showed variations across the business.
- Different product subcategories contributed differently to overall profitability.

- **Provide the analysis insights:**

Descriptive Analysis

- Analyse total sales, net sales, total profit, and quantity sold.
- Identify monthly sales trends and top-performing product categories.
- Compare sales and profitability across different products and regions.
- Measure the overall product return rate.

Diagnostic Analysis

- Identify factors contributing to high or low sales and profitability.
- Analyse how discounts affect net sales and profit margins.
- Investigate which product categories generate lower profits.
- Examine product return patterns to identify areas for improvement.

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10. Conclusions

The integration of Python and Power BI proved effective in performing end-to-end retail data analysis, from data cleaning and preprocessing to exploratory data analysis and interactive dashboard development.

- Python was used to clean, transform, and analyse retail sales data to identify meaningful trends and patterns.
- Power BI enabled the creation of two interactive dashboards focusing on sales performance, profitability, customer behaviour, and product analysis.
- The analysis provided insights into regional sales performance, product category trends, customer satisfaction, discounts, and product returns.
- Identified key business patterns that can help improve sales strategies, customer retention, and profitability.
- The project demonstrated how data analytics and visualization can transform raw retail data into meaningful business insights and support data-driven decision-making.

Overall Conclusion:

This project enhanced my practical knowledge of Python-based data analysis and Power BI dashboard development while demonstrating the importance of data-driven insights in understanding retail business performance and supporting effective business decisions.